Md Shaffaf Hussain 108

a) Loading the dataset into a Panda DataFrame

```
import pandas as pd

data = {
    'ID':[1,2,3,4,5,6],
    'Name':['Rahul','Priya','Aman','Neha','Rahul','Karan'],
    'Age':[25,30,35,28,25,40],
    'Salary':[30000,40000,None,35000,30000,120000],
    'Department':['IT','HR','IT','Finance','IT','IT'],
    'Gender':['M','F','M','F','M','M'],
    'Experience':[2,5,8,4,2,15]
}

df = pd.DataFrame(data)
print(df)
```

```
   ID   Name  Age    Salary Department Gender  Experience
0   1  Rahul   25   30000.0         IT      M           2
1   2  Priya   30   40000.0         HR      F           5
2   3   Aman   35       NaN         IT      M           8
3   4   Neha   28   35000.0    Finance      F           4
4   5  Rahul   25   30000.0         IT      M           2
5   6  Karan   40  120000.0         IT      M          15
```

Handling the missing value of the Salary

```
df['Salary'].fillna(df['Salary'].mean())
```

	Salary
0	30000.0
1	40000.0
2	51000.0
3	35000.0
4	30000.0
5	120000.0

dtype: float64

Apply Min-Max normalization to Salary

```
from sklearn.preprocessing import MinMaxScaler
```

```
scaler = MinMaxScaler()
df['Salary_MinMax'] = scaler.fit_transform(df[['Salary']])
df
```

	ID	Name	Age	Salary	Gender	Experience	Department_Finance	Department_HR	Department_IT	Salary_MinMax
0	1	Rahul	25	30000.0	M	2	False	False	True	0.000000
1	2	Priya	30	40000.0	F	5	False	True	False	0.111111
2	3	Aman	35	NaN	M	8	False	False	True	NaN
3	4	Neha	28	35000.0	F	4	True	False	False	0.055556
4	5	Rahul	25	30000.0	M	2	False	False	True	0.000000
5	6	Karan	40	120000.0	M	15	False	False	True	1.000000

Next steps: Generate code with df New interactive sheet

Convert Department column using One-Hot Encoding

```
df = pd.get_dummies(df, columns=['Department'])

df
```

	ID	Name	Age	Salary	Gender	Experience	Department_Finance	Department_HR	Department_IT
0	1	Rahul	25	30000.0	M	2	False	False	True
1	2	Priya	30	40000.0	F	5	False	True	False
2	3	Aman	35	NaN	M	8	False	False	True
3	4	Neha	28	35000.0	F	4	True	False	False
4	5	Rahul	25	30000.0	M	2	False	False	True
5	6	Karan	40	120000.0	M	15	False	False	True

Next steps: Generate code with df New interactive sheet

Apply Z-score Standardization to Age

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()

df['Age_Zscore'] = scaler.fit_transform(df[['Age']])

df
```

	ID	Name	Age	Salary	Gender	Experience	Department_Finance	Department_HR	Department_IT	Age_Zscore
0	1	Rahul	25	30000.0	M	2	False	False	True	-1.011205
1	2	Priya	30	40000.0	F	5	False	True	False	-0.091928
2	3	Aman	35	NaN	M	8	False	False	True	0.827349
3	4	Neha	28	35000.0	F	4	True	False	False	-0.459639
4	5	Rahul	25	30000.0	M	2	False	False	True	-1.011205
5	6	Karan	40	120000.0	M	15	False	False	True	1.746627

Next steps: Generate code with df New interactive sheet

```
print(df)
```

```
   ID   Name  Age    Salary Gender  Experience  Department_Finance  \
0   1  Rahul   25   30000.0      M           2               False
1   2  Priya   30   40000.0      F           5               False
2   3   Aman   35       NaN      M           8               False
3   4   Neha   28   35000.0      F           4                True
4   5  Rahul   25   30000.0      M           2               False
5   6  Karan   40  120000.0      M          15               False

   Department_HR  Department_IT  Age_Zscore
0          False           True   -1.011205
1           True          False   -0.091928
2          False           True    0.827349
3          False          False   -0.459639
4          False           True   -1.011205
5          False           True    1.746627
```